

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

2. Radio Wave Propagation in Mobile Network Environment

12 hrs, **the largest chapter in the syllabus** · asked in 22 of 22 papers · 7 syllabus sub-topics · **by far the heaviest numerical load** → Ch 3 numerical companion.

Read this first:

- The chapter splits cleanly in two, and the exam treats them separately.
 - **Large-scale path loss:** how far the signal reaches, averaged over hundreds of wavelengths. Free space, two-ray, Okumura, Hata, indoor.
 - **Small-scale fading:** how the signal flutters over a few wavelengths. Delay spread, Doppler, coherence bandwidth, Rayleigh / Rician.
- Three questions are near-certain: **small-scale fading types + factors** **TOP 6/22**, **coherence bandwidth and coherence time** **HOT 5/22**, and an **Okumura or Hata numerical** **HOT 4/22**.
- Two derivations are worth memorising line by line, because they recur: the **two-ray path difference and phase difference** **HOT 4/22** and the **Doppler shift** **HOT 3/22**.
- **Nearly every question here has a numerical twin.** Learn the theory with the companion open.

3.1 Free space propagation

Explain free space propagation. Relate radiated power to electric field **8** **2+2+2+2** **6**

HOT 3/22 80 Ba, 81 Ba, 78 Ch, 80 Bh · **almost always set as a numerical** → Ch 3 companion

When it applies

- Used to predict received signal strength when Tx and Rx have a **clear, unobstructed LOS path**.
- Eg satellite links, microwave LOS links.
- **Almost never true for a mobile in a city**, but every other model is built as a correction to it.

Step 1: power flux density

- An **isotropic** source radiates P_t uniformly over a sphere of area $4\pi d^2$.
- **Power flux density** at distance d :

$$P_d = \frac{P_t G_t}{4\pi d^2} \quad \text{W/m}^2$$

- **EIRP** = $P_t G_t$ watts: the power an isotropic antenna would need to give the same reading in the main beam.
- **ERP** uses a half-wave dipole as the reference instead: $\text{ERP} = \text{EIRP} - 2.15 \text{ dB}$.

Step 2: effective aperture

- **Effective aperture** A_e = ratio of power available at the antenna terminals to the incident power per unit area.

$$A_e = \frac{G_r \lambda^2}{4\pi}, \quad \lambda = \frac{c}{f}$$

Step 3: the Friis equation

- $P_r = P_d \cdot A_e$, so

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

- $L \geq 1$ = system loss (feeder attenuation, filters, mismatch). $L = 1$ means no hardware loss.
- $P_r \propto 1/d^2$ **and** $\propto \lambda^2$: doubling distance costs 6 dB, doubling frequency costs 6 dB.

Relating power to E-field (the 78 Ch / 80 Ba style numerical)

- Power flux density in terms of the field:

$$P_d = \frac{|E|^2}{\eta} = \frac{|E|^2}{120\pi}, \quad \eta = 377 \, \Omega \text{ (free space)}$$

- So E falls as $1/d$ while power falls as $1/d^2$.
- Received power through the aperture:

$$P_r(d) = \frac{|E|^2}{120\pi} \cdot \frac{G_r \lambda^2}{4\pi}$$

- **rms voltage** into a matched receiver of input resistance R_{ant} :

$$P_r = \frac{V_{\text{rms}}^2}{R_{\text{ant}}} = \frac{(V_{\text{ant}}/2)^2}{R_{\text{ant}}} \Rightarrow V_{\text{ant}} = 2\sqrt{P_r R_{\text{ant}}}$$

The factor 2 is the matched-load voltage divider. Losing it is the standard mistake.

Path loss

- **With** antenna gains:

$$PL(\text{dB}) = 10 \log \frac{P_t}{P_r} = -10 \log \left[\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2} \right]$$

- **Without** antenna gains ($G_t = G_r = 1$):

$$PL(\text{dB}) = -10 \log \left[\frac{\lambda^2}{(4\pi)^2 d^2} \right]$$

- Useful dB form, d in km and f in MHz:

$$PL_{\text{FS}} = 32.44 + 20 \log f_{\text{MHz}} + 20 \log d_{\text{km}}$$

verified/added

- The minus sign on the gain term says that term is a **gain**. The $(4\pi d/\lambda)^2$ term is the **free space loss** proper.
- Raising d raises the loss, which pushes designers to **lower frequencies** to keep the link budget manageable.

Far field (Fraunhofer region)

- Friis is valid only **beyond** the far-field distance

$$d_f = \frac{2D^2}{\lambda}$$

where D = the largest physical linear dimension of the antenna.

- Also needs $d_f \gg D$ and $d_f \gg \lambda$.
- The equation blows up at $d = 0$, so a **close-in reference distance** d_0 is used:

$$P_r(d) = P_r(d_0) \left(\frac{d_0}{d} \right)^2, \quad d \geq d_0 \geq d_f$$

- d_0 is typically **1 km** for large cells, **100 m** for microcells, **1 m** indoors.

3.2 The three propagation mechanisms

Explain the three basic radio wave propagation mechanisms. Large scale Vs small scale model 3+6 3

8

HOT 3/22 72 Ash, 73 Bh, 72 Ma

- Apart from the direct wave, the receiver gets **reflected**, **diffracted** and **scattered** waves.
- They add **vectorially**, so the resultant varies in strength in real time. That is the physical origin of fading.

1. Reflection

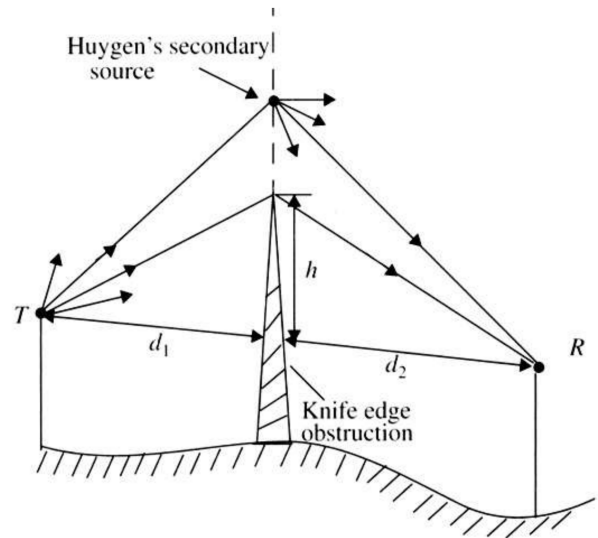
- Wave hits an object **very large compared to the wavelength**.
- Eg the earth's surface, buildings, walls.
- Part reflects, part transmits into the object.
- Strength set by the **reflection coefficient** Γ , which depends on polarisation, angle of incidence and the material.
- At grazing incidence $\Gamma \rightarrow -1$, ie **full reflection with a 180° phase flip**. That is the assumption the two-ray model rests on.

2. Diffraction (79 Bh pairs this with a Fresnel derivation, 2+6)

- Path is obstructed by a surface with **sharp edges**.
- Secondary waves bend around the obstacle into the shadow region.
- **This is what lets a mobile work with no LOS at all** in urban and hilly terrain.
- Explained by **Huygens' principle**: every point on a wavefront acts as a point source of secondary wavelets, which add to form the new front.
- Diffraction loss grows as the obstruction fills more of the first Fresnel zone.

3. Scattering

- Medium contains objects **small compared to the wavelength**, or rough surfaces.
- Eg foliage, street signs, lamp posts, rain drops, snow.
- Energy is **spread in many directions**.
- Follows the same principles as diffraction.
- **Received signal is often stronger than reflection and diffraction alone predict**, because scattering fills in the gaps.



Knife-edge diffraction geometry. Effective obstruction height h , distances d_1 and d_2 from Tx and Rx.

Fresnel zones and the diffraction parameter

- A **Fresnel zone** is the ellipsoid of revolution over which secondary waves reach the receiver with an excess path of $n\lambda/2$.
- Numbered $F_1, F_2, F_3, \dots$. Odd zones add constructively, even zones destructively.
- Size set by **frequency** and **Tx-Rx distance**.
- **Fresnel-Kirchhoff diffraction parameter**

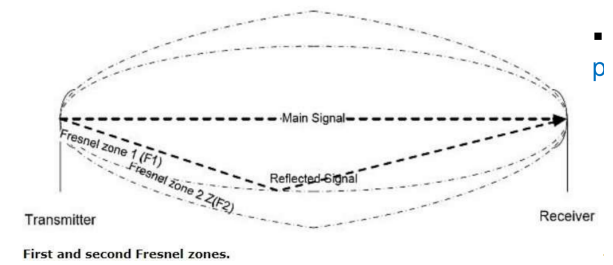
$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}}$$

- Radius of the n -th zone at distance d_1 :

$$r_n = \sqrt{\frac{n\lambda d_1 d_2}{d_1 + d_2}}$$

verified/added

- **Rule of thumb: keep 55 % of the first Fresnel zone clear** and the link behaves as free space.
- Diffraction gain G_d (dB) against v :
 - $v \leq -1$: $G_d = 0$
 - $-1 \leq v \leq 0$: $G_d = 20 \log(0.5 - 0.62v)$
 - $0 \leq v \leq 1$: $G_d = 20 \log(0.5 e^{-0.95v})$
 - $v > 2.4$: $G_d = 20 \log(0.225/v)$



First and second Fresnel zones.

First and second Fresnel zones.

3.3 Two-ray ground reflection model

Derive the path difference and phase difference in the two-ray model, and the path loss 8 2+6 2+8 6

HOT 4/22 71 Bh, 74 Bh, 76 Bh, 81 Bh · 76 Bh asks first for the (+) over free space, so open with that

Why it beats the free space model (the 2-mark opener)

- Free space assumes **one** path. A real mobile link always has a **ground-reflected** path as well.
- The two-ray model adds that second path, so it predicts the $1/d^4$ **roll-off actually measured** over large distances, instead of $1/d^2$.
- Found **reasonably accurate** for large-cell mobile systems over a few km.
- Needs no measured data, unlike Okumura or Hata.

Assumptions

- $h_t, h_r \gg \lambda$ and $h_t, h_r \ll d$.
- Reflected wave arrives at **grazing incidence**, so $\Gamma = -1$.
- Flat earth, no curvature.

Step 1: path difference Δ (method of images)

- Replace the reflection by an **image transmitter** at $-h_t$.
- LOS path and reflected path become two straight lines:

$$d' = \sqrt{(h_t - h_r)^2 + d^2}, \quad d'' = \sqrt{(h_t + h_r)^2 + d^2}$$

- So

$$\Delta = d'' - d' = d\sqrt{1 + \left(\frac{h_t + h_r}{d}\right)^2} - d\sqrt{1 + \left(\frac{h_t - h_r}{d}\right)^2}$$

- Expand each root binomially, valid since $d \gg h_t, h_r$:

$$\Delta \approx d \left\{ \left(1 + \frac{1}{2} \left(\frac{h_t + h_r}{d}\right)^2\right) - \left(1 + \frac{1}{2} \left(\frac{h_t - h_r}{d}\right)^2\right) \right\}$$

$$\Delta \approx \frac{2h_t h_r}{d}$$

Step 2: phase difference and time delay

$$\theta_\Delta = \frac{2\pi\Delta}{\lambda} = \frac{4\pi h_t h_r}{\lambda d}$$

$$\tau_d = \frac{\Delta}{c} = \frac{\theta_\Delta}{2\pi f_c}$$

Step 3: total E-field

- For large d , $|E_0 d_0 / d'| \approx |E_0 d_0 / d''| \approx |E_0 d_0 / d|$, so the two rays differ **only in phase**:

$$|E_{\text{tot}}(d)| = \frac{E_0 d_0}{d} |\cos \theta_\Delta - \cos 0|$$

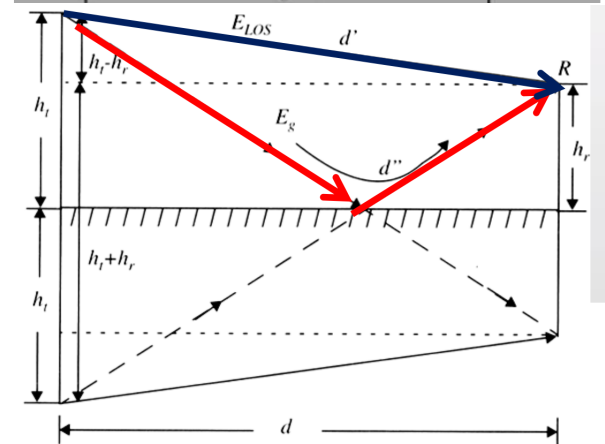
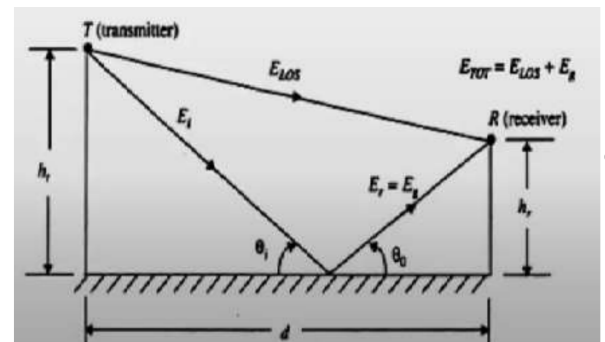
- Using $\cos \alpha - \cos \beta$ identity:

$$|E_{\text{tot}}(d)| = 2 \frac{E_0 d_0}{d} \sin\left(\frac{\theta_\Delta}{2}\right)$$

- For $\theta_\Delta/2 < 0.3$ rad, $\sin x \approx x$:

$$|E_{\text{tot}}(d)| \approx \frac{2E_0 d_0}{d} \cdot \frac{2\pi h_t h_r}{\lambda d} = \frac{k}{d^2}$$

- The field now falls as $1/d^2$, so power falls as $1/d^4$.**



Method of images: the ground-reflected ray is redrawn as a straight line from the image transmitter at $-h_t$.

Step 4: received power and path loss

- Using $P_r = |E_{\text{tot}}|^2 A_e / \eta$:

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$$

- Note λ has cancelled:** the two-ray result is **independent of frequency**.
- In dB:

$$PL(\text{dB}) = 40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

- Valid only beyond the **crossover distance**
 $d_c = 4\pi h_t h_r / \lambda$; below it the model oscillates and free space is used instead. verified/added

3.4 Practical link budget design

Explain the log-distance path loss model and log-normal shadowing 4 2+4

PIN 1/22 77 Ch · the **link budget numericals** built on it are 10 and 12 marks (74 Bh, 76 Bh) → Ch 3 companion

Log-distance path loss model

- Both theory and measurement show received power falls **logarithmically** with distance, indoors and outdoors.
- Generalises free space by letting the exponent be n instead of 2:

$$\overline{PL}(d) \propto \left(\frac{d}{d_0}\right)^n$$

$$\overline{PL}(d) [\text{dB}] = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right)$$

- $\overline{PL}(d_0)$ is computed by the **free space** formula, or measured.
- Bars mean **ensemble average** over all possible paths at that distance.
- One number, n , absorbs the whole environment.**

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

Log-normal shadowing

- Problem: two locations at the **same** d can have wildly different surroundings, so a single average is wrong for any one site.
- Measurement shows $PL(d)$ at a given d is **random**, and **log-normally** distributed about the average.
- Add a zero-mean Gaussian random variable X_σ in dB, standard deviation σ dB:

$$PL(d) = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma$$

$$P_r(d) [\text{dBm}] = P_t [\text{dBm}] - PL(d)$$

- Called **shadowing**: the mobile is shadowed by clutter (buildings, hills, foliage).
- n and σ are found from measured data by **linear regression**, minimising mean square error.
- Small σ = the model fits that environment well.**

Link budget design

- A link budget adds every gain and subtracts every loss between transmitter output and receiver input, then compares against the receiver **sensitivity**.
- Generic form, all terms in dB:

$$P_r = P_t + G_t + G_r - L_{\text{cable,t}} - L_{\text{cable,r}} - L_{\text{misc}} - PL(d)$$

- Link closes if $P_r \geq \text{sensitivity} + \text{link margin}$.
- Two** budgets are needed, forward and reverse, and the **weaker one sets the cell radius**.

How to run the exam version

- Compute **maximum allowable path loss** on each link:

$$PL_{\text{max}} = P_t + G_t + G_r - L_{\text{cables}} - \text{sensitivity} - \text{margin}$$

- Put PL_{max} into the propagation model given (usually **Okumura**) and solve for d .
- Take the **smaller** of the forward and reverse distances.
- Compare against the range asked for and state whether the link is feasible.

Two ways to build a propagation model

- Empirical**: collect measurements, fit curves. Eg Okumura, Hata.
 - Accurate in the environment measured, no guarantee elsewhere.
- Analytical**: model the physics, derive equations. Eg free space, two-ray.
 - Purely analytical models are inaccurate because so many assumptions are needed to keep the derivation tractable.
- Practical models are hybrids**: an analytical core with measured correction factors.

3.5 Outdoor propagation models

Explain any two outdoor propagation models. State the conditions for Okumura and Hata 3+3 5 2+2
4

TOP 7/22 71 Ma, 73 Ma, 75 Bh, 79 Ch, 80 Ch, 81 Bh, 80 Bh · the numerical version appears in 13 of 22 papers → Ch 3 companion

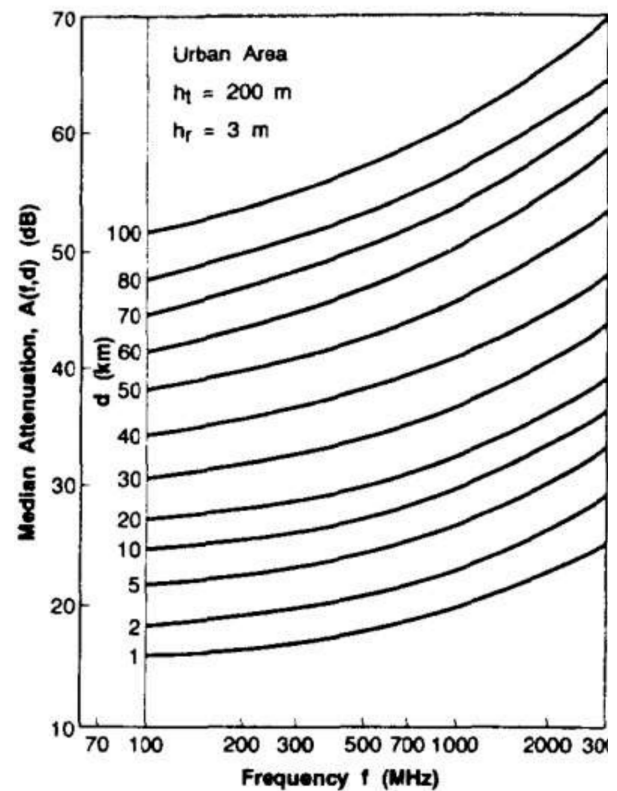
Outdoor transmission runs over irregular terrain, so the **terrain profile**, trees, buildings and hills must all be accounted for. Purely analytical models cannot do this, so all the outdoor models below are **empirical**.

Okumura model 4

- **Wholly empirical.** Built from extensive drive-test measurements in and around Tokyo, 1968.
- **The single most widely used model for signal prediction in urban areas.**
- Okumura published a set of **curves** of median attenuation relative to free space, $A_{mu}(f, d)$, for an urban area over quasi-smooth terrain, plus correction curves.
- **Validity conditions (the 2-mark answer)**
 - Frequency **150 MHz to 1920 MHz** (extrapolated to 3 GHz).
 - Distance **1 km to 100 km**.
 - BS antenna height **30 m to 1000 m**.
 - Quasi-smooth terrain, urban reference.
- **The formula**

$$L_{50}(\text{dB}) = L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{\text{AREA}}$$

- L_{50} = median (50th percentile) path loss,
- L_F = free space path loss,
- A_{mu} = median attenuation relative to free space, read off the curve,
- $G(h_{te}), G(h_{re})$ = antenna height gain factors,
- G_{AREA} = gain from the environment type.



$A_{mu}(f, d)$ over a quasi-smooth urban area, $h_{te} = 200$ m,
 $h_{re} = 3$ m.

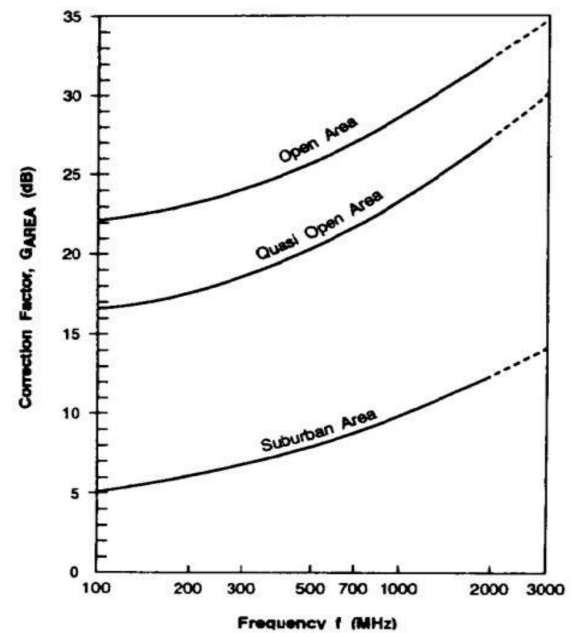
– Height gain factors:

$$G(h_{te}) = 20 \log \frac{h_{te}}{200}, \quad 30 \text{ m} < h_{te} < 1000 \text{ m}$$

$$G(h_{re}) = 10 \log \frac{h_{re}}{3}, \quad h_{re} \leq 3 \text{ m}$$

$$G(h_{re}) = 20 \log \frac{h_{re}}{3}, \quad 3 \text{ m} < h_{re} < 10 \text{ m}$$

- (+) simple, and **the most accurate model available for mature cellular and land-mobile systems in cluttered urban areas.**
- (–) slow response to rapid terrain change → good in cities, poorer in rural areas. Deviation from measurement is 10 to 14 dB.
- (–) needs the curves, so it cannot be evaluated from formulas alone.



G_{AREA} for suburban, quasi-open and open areas.

Hata model 4 6 7

- **Hata is Okumura's curves turned into formulas**, so it can be worked by hand.
- Same measured data, so the same physical assumptions.
- **Validity conditions**
 - Frequency **150 MHz to 1500 MHz**.
 - Distance **1 km to 20 km**.
 - BS antenna **30 m to 200 m**, MS antenna **1 m to 10 m**.
 - **No path-specific correction** is possible, unlike Okumura.
- **Urban median path loss:**

$$L_{50}(\text{urban}) = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$$

with f_c in MHz, h_{te} , h_{re} in m, d in km.

- **Mobile antenna correction $a(h_{re})$:**
 - small / medium city: $(1.1 \log f_c - 0.7)h_{re} - (1.56 \log f_c - 0.8)$ dB
 - large city, $f_c \leq 300$ MHz: $8.29(\log 1.54h_{re})^2 - 1.1$ dB
 - large city, $f_c > 300$ MHz: $3.2(\log 11.75h_{re})^2 - 4.97$ dB
- **Suburban:** $L_{50}(\text{urban}) - 2[\log(f_c/28)]^2 - 5.4$
- **Open rural:** $L_{50}(\text{urban}) - 4.78(\log f_c)^2 + 18.33 \log f_c - 40.98$
- Suburban and rural terms are just **reductions in fixed loss** for less cluttered environments.

Okumura Vs Hata (81 Bh asks exactly this, 4)

- **Form:** Okumura is graphical, Hata is closed-form.
- **Frequency:** Okumura to 1920 MHz, Hata to 1500 MHz.
- **Distance:** Okumura 1 to 100 km, Hata 1 to 20 km.
- **BS height:** Okumura 30 to 1000 m, Hata 30 to 200 m.
- **Path-specific corrections:** Okumura ✓, Hata ×
- **Accuracy:** within **1 dB of Okumura for $d > 1$ km**, so Hata is preferred for large-cell work. Okumura is better for small cells and irregular terrain.
- **Extension:** COST-231 extends Hata to **2 GHz** by changing the constants, for PCS bands. verified/added

Longley–Rice (ITS irregular terrain model)

- Point-to-point, **40 MHz to 100 GHz**, over irregular terrain.
- Runs as a **computer program**. Inputs: frequency, path length, polarisation, antenna heights, surface refractivity, ground conductivity and permittivity, climate.
- Two modes:
 - **point-to-point**, using a detailed terrain profile,
 - **area**, using empirical estimates where no profile exists.
- (–) **ignores multipath, buildings and foliage**, so it is a terrain model, not an urban model.

Walfisch–Bertoni syllabus, no PYQ yet

- Models **rooftop diffraction** explicitly, so it fits dense urban areas where the BS antenna is above the rooftops.
- Path loss is the product of three factors:

$$S = P_0 Q^2 P_1$$

- P_0 = free space loss between isotropic antennas,
- Q^2 = reduction from the **row of buildings** shadowing the street,
- P_1 = loss from the **final rooftop-to-street** diffraction.
- Predicts an overall $\propto d^{-3.8}$ roll-off, matching urban measurement.
- Adopted as part of **COST-231**.

Ericsson multiple breakpoint model (80 Bh asks by name, 2)

- Built from measurements in a multi-floor office building.
- Uses a **uniform distribution** to generate path loss between a minimum and maximum bound at each distance.
- **4 breakpoints**, each carrying an upper and lower bound on path loss.
- Assumes **30 dB attenuation** at $d_0 = 1$ m, accurate at 900 MHz w/ unity-gain antennas.
- Slope changes from **20 dB/decade** at the first breakpoint to **60 dB/decade** in the final section.
- **Its value is that it gives a deterministic limit on the range of path loss**, not a single number.

3.6 Indoor propagation models**Explain indoor propagation models (any two). What factors influence them?** 8 5

PIN 2/22 70 Bh, 82 Bh

How indoor differs from outdoor (the “factors” answer)

- Distances covered are **much smaller**.
- Environment varies **far more** over a much smaller Tx–Rx separation.
- Propagation is governed by the **same three mechanisms**, but conditions are far more variable.
- **Factors that influence it:**
 - **Layout** of the building, room sizes, corridors.
 - **Construction materials:** concrete, metal, plaster, glass.
 - **Building type:** office, residential, factory.
 - Whether doors are **open or closed**.
 - Where the antenna is mounted: **desk height Vs ceiling**.
 - Number of **floors** between Tx and Rx.
 - Number of **windows** and whether they are tinted (metallised tint is a strong reflector).
 - Movement of **people and furniture**.
- **Path loss exponents indoors range from 1.6 (LOS corridor, better than free space, because of waveguiding) to 6 (obstructed).**

Model 1: log-distance path loss (same form as outdoor)

$$PL(d) [\text{dB}] = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma$$

- n and σ are tabulated **per building type**.
- Smaller $\sigma \rightarrow$ better accuracy.
- d_0 is typically **1 m** indoors.

Model 2: attenuation factor model

$$PL(d) [\text{dB}] = \overline{PL}(d_0) + 10 n_{SF} \log\left(\frac{d}{d_0}\right) + \text{FAF} + \sum \text{PAF}$$

- n_{SF} = path loss exponent measured on the **same floor**.
- **FAF** = floor attenuation factor, added once per floor penetrated.
- **PAF** = partition attenuation factor, summed over the partitions crossed.
- **Standard deviation drops to about 4 dB**, against 13 dB for the plain log-distance model.

Partition losses

- **Hard partitions:** structural walls of a room. Fixed.
- **Soft partitions:** movable dividers that do not reach the ceiling.
- **Between floors:** FAF is **not linear in floor count**. The first floor costs the most (about 15 dB), each further floor adds only 6 to 10 dB, then it saturates.
- Reason: signal also travels **out of and back into** the building via windows, so the direct path stops dominating.

Material Type	Loss (dB)	Frequency
All metal partition	26 dB	815 MHz
Concrete block wall	13 dB	1300 MHz
Empty Cardboard boxes	3-6 dB	1300 MHz
Dry Plywood (0.75 inches)	1 dB	9.6 GHz
Dry Plywood (0.7 inches)	4 dB	28.2 GHz

Measured same-floor partition losses.

3.7 Small-scale fading and multipath

What is small-scale fading? Give its types and the factors that influence it 2+4+4 4+4 3+6 1+2+5 2+2

TOP 6/22 72 Ma, 73 Ma, 75 Bh, 78 Ch, 80 Ba, 81 Ba · the most repeated question in the chapter

Definition

- **Small-scale fading** = rapid fluctuation of the amplitude, phase and multipath delay of a radio signal over a **short travel distance or short time interval**.
- Caused by **interference between two or more copies** of the transmitted signal that arrive at slightly different times.
- The copies (multipath waves) add **vectorially** with random phases, so the sum behaves like noise.
- **Received power can change by 30 to 40 dB when the receiver moves a fraction of a wavelength.**
- Large-scale path loss can be ignored over such a short distance.

The three effects it produces

- **Rapid changes in signal strength** over a small travel distance or time interval.
- **Random frequency modulation** due to different **Doppler shifts** on different multipath components.
- **Time dispersion (echoes)** caused by multipath propagation delays → **ISI**.

Time dispersion Vs frequency dispersion

- **Time dispersion:** received signal **lasts longer** than the transmitted one, because paths have different delays. Measured by **delay spread**.
- **Frequency dispersion:** received signal is **wider in frequency** than the transmitted one, because paths have different Doppler shifts. Measured by **Doppler spread**.
- Both distort the received signal, and they are **independent** of each other. **That is why fading is classified twice over, once by each.**

Impulse response model of the multipath channel

syllabus, no PYQ yet

- The channel is treated as a **linear filter with a time-varying impulse response**.
- Feed it a unit impulse $\delta(t)$ and the output is

$$h(t) = \sum_{n=1}^N A_n \exp(-j\phi_n) \delta(t - \tau_n)$$

- A_n, τ_n, ϕ_n = amplitude, arrival delay and phase of the n -th path.
- Time variation comes from motion, so it is really $h(t, \tau)$: τ indexes the multipath delay, t indexes the mobile's position.
- **This one model generates everything else in the chapter:** the PDP is $|h(t, \tau)|^2$, and the delay-spread parameters are its moments.

Factors influencing small-scale fading (learn all 4)

- **Multipath propagation.**
 - Random amplitude and phase of each component → the sum fluctuates.
 - Also **lengthens** the time the baseband signal takes to arrive → signal smearing → ISI.
- **Speed of the mobile.**
 - Relative motion causes a **Doppler shift** f_d on each component.
 - Different components arrive from different angles → different shifts → random FM.
- **Speed of surrounding objects.**
 - If nearby objects move **faster than the mobile**, they dominate the fading.
 - If they move slower, ignore them.
- **Transmission bandwidth of the signal.**
 - If signal bandwidth $>$ channel bandwidth, the received signal is **distorted**, but its strength does not fade much over a local area.
 - If signal bandwidth $<$ channel bandwidth, amplitude **changes rapidly** but the signal is not distorted in time.
 - **A channel gives you distortion or fading, typically one or the other.**

Multipath measurement techniques syllabus, no PYQ yet

- **Direct RF pulse sounding:** transmit a narrow pulse, display the received power against delay on a scope. Simple, but a single pulse can be lost in a fade.
- **Spread spectrum sliding correlator:** transmit a PN sequence, correlate at the receiver w/ a slightly slower clock so the correlation peak slides through the delays. Good dynamic range, rejects interference.
- **Swept frequency (frequency domain) sounding:** sweep a network analyser across the band, measure S_{21} , then inverse Fourier transform to get $h(\tau)$. Needs a cabled link, so it suits indoor work only.

Define Doppler shift and Doppler spread. Derive the expression for Doppler shift 2+4 1+2+5 2

HOT 4/22 78 Ch, 81 Ba, 80 Ch, 82 Bh · 82 Bh words it as "significance of frequency dispersion"

Derivation (reproduce these 5 lines exactly)

- Mobile moves at constant velocity v along a segment of length d from X to Y , receiving from a **distant** source S .
- Source is far away → the arrival angle θ is the same at X and Y .
- **Path length difference**

$$\Delta l = d \cos \theta = v \Delta t \cos \theta$$

- **Phase change** in the received signal

$$\Delta \phi = \frac{2\pi \Delta l}{\lambda} = \frac{2\pi v \Delta t}{\lambda} \cos \theta$$

- **Apparent frequency change** is the rate of change of phase:

$$f_d = \frac{1}{2\pi} \cdot \frac{\Delta \phi}{\Delta t} \Rightarrow \boxed{f_d = \frac{v}{\lambda} \cos \theta}$$

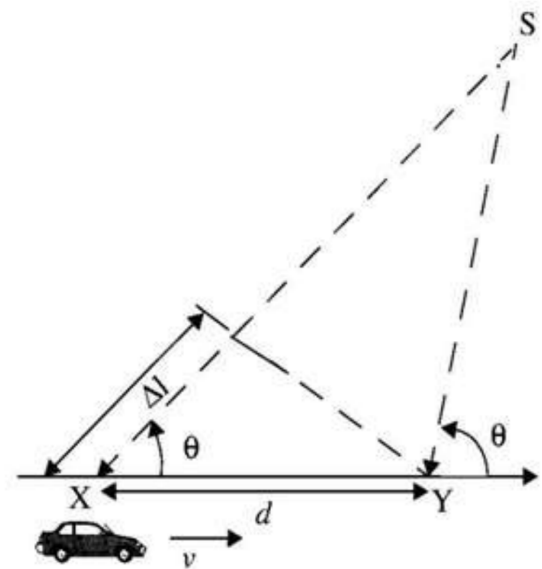


Illustration of Doppler effect.

Mobile moving from X to Y at velocity v ; source S is far away, so θ is unchanged.

Reading the result

- $\theta = 0^\circ$, moving **towards** the source: $f_d = +v/\lambda$, maximum **positive** shift. Received frequency $f_c + f_d$.
- $\theta = 180^\circ$, moving **away**: $f_d = -v/\lambda$, maximum negative shift.
- $\theta = 90^\circ$, moving **perpendicular**: $f_d = 0$, no shift at all.
- **Maximum Doppler shift**

$$f_m = \frac{v}{\lambda} = \frac{vf_c}{c}$$

Doppler spread B_D

- Measure of the **spectral broadening** caused by the time rate of change of the channel.
- Defined as the **maximum Doppler shift**:
 $B_D = f_m = v/\lambda$.
- A pure tone at f_c comes out spread over $f_c \pm f_m$.
- Characterises the channel's **frequency dispersiveness**.

Significance of frequency dispersion (82 Bh)

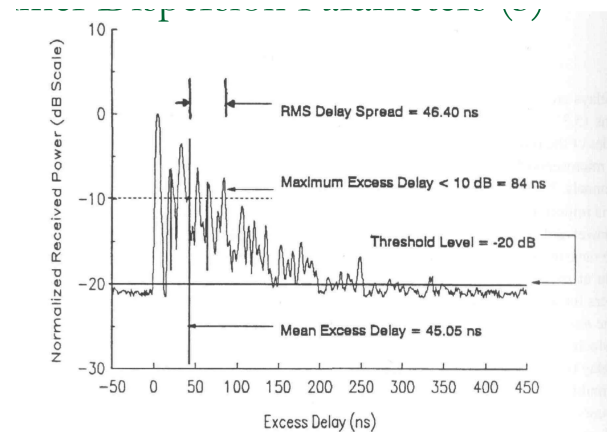
- It makes the channel **time-varying**, so the receiver must keep re-estimating it.
- If B_D approaches the signal bandwidth, the channel changes **within one symbol** → **fast fading, and an irreducible BER floor that more transmit power cannot cure**.
- It sets how often an **adaptive equalizer must retrain** and how fast a **carrier recovery loop** must track.
- Different multipath components have different f_d , so the composite signal suffers **random FM**, heard as noise on an analog system.

Explain the parameters of the mobile multipath channel 7 4 2+4

HOT 3/22 71 Ma, 80 Bh, 70 Ma · 80 Bh asks which parameters are used to **classify the types of fading** · **numericals on this appear in 5 papers** → Ch 3 companion

Power delay profile (PDP)

- **PDP** = received power against **excess delay**, ie the intensity of the signal arriving through the multipath channel as a function of delay.
- Obtained by **averaging many impulse response measurements** over a local area.
- Every time-dispersion parameter below is a **moment of the PDP**.



A measured PDP w/ its mean excess delay, rms delay spread and 10 dB maximum excess delay marked. Note the -20 dB noise threshold.

Time dispersion parameters

- **First arrival delay** τ_A : delay of the first signal to reach the receiver, set by the shortest path. Used as the **reference**, so every other delay is an excess delay.
- **Mean excess delay** = first moment of the PDP:

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k}{\sum_k P(\tau_k)}$$

- **rms delay spread** = square root of the second central moment:

$$\sigma_\tau = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}, \quad \overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2}$$

- **Maximum excess delay (X dB)** = $\tau_X - \tau_0$, the delay over which multipath energy stays within X dB of the strongest component.
- σ_τ **is the one that matters in practice**, because it, not $\tau_{\max}$, predicts ISI.
- These values **depend on the noise threshold** chosen. Set the threshold too low and noise is counted as multipath, inflating every parameter.

Coherence bandwidth B_c

- **Definition**: the range of frequencies over which the channel passes all spectral components with **approximately equal gain and linear phase**, ie over which the channel looks **flat**.
- It is the **frequency-domain dual** of the rms delay spread. There is no exact relationship, only rules of thumb:

$$B_c \approx \frac{1}{50 \sigma_\tau} \quad (0.9 \text{ correlation, "90 \%"})$$

$$B_c \approx \frac{1}{5 \sigma_\tau} \quad (0.5 \text{ correlation, "50 \%"})$$

- Two sinusoids separated by **more** than B_c are affected **differently** by the channel.
- **Significance**: compare B_c against the signal bandwidth to decide whether the channel is **flat or frequency selective**. That single comparison decides whether an equalizer is needed.

Coherence time T_c

- **Definition**: the time over which the channel impulse response is essentially **time-invariant**.
- Dual of the Doppler spread:

$$T_c \approx \frac{1}{f_m}, \quad T_c \approx \sqrt{\frac{9}{16\pi f_m^2}} = \frac{0.423}{f_m}$$

The second (geometric mean) form is the one normally used.

- Two signals arriving more than T_c apart are affected **differently**.
- **Large** $T_c \rightarrow$ channel changes slowly.
- **Significance**: compare T_c against the symbol period to decide whether the channel is **slow or fast fading**. Also sets how often the equalizer must retrain.

Explain the types of small-scale fading 4+4 1+3 2+7 2+4

TOP 6/22 70 Ma, 74 Ma, 79 Ch, 81 Ba, 80 Ch, 80 Bh · fading is classified twice over: once by delay spread, once by Doppler spread. Always give both tables.

A. By multipath time delay spread

	Flat fading	Frequency selective
Bandwidth	$B_S \ll B_C$	$B_S > B_C$
Delay spread	$\sigma_\tau \ll T_S$	$\sigma_\tau > T_S$
Spectrum of the signal	preserved	not preserved
Channel acts as	flat gain, linear phase	a filter
ISI	none	severe
Cure	raise transmit power, use diversity	equalizer

- **Flat fading** is the most common type.
- Signal bandwidth is narrow, so the whole signal rides up and down together.
- Causes **deep fades**, so flat-fading channels are sometimes called **amplitude varying** or **narrowband** channels.
- **Frequency selective fading** occurs when the delayed copies overlap **into the next symbol**.
- **Frequency selective fading is harder to model**, because the channel response depends on the exact multipath structure.

B. By Doppler spread

	Fast fading	Slow fading
Doppler spread	high, $B_S < B_D$	low, $B_S \gg B_D$
Coherence time	$T_C < T_S$	$T_C \gg T_S$
Channel changes	within a symbol	over many symbols
Relative rates	channel varies faster than the baseband signal	channel varies slower than the baseband signal

- Fast / slow says **nothing** about flat / selective. **The four combinations are all possible**: flat slow, flat fast, frequency selective slow, frequency selective fast.
- **Fast fading** arises from motion, or from motion of the surroundings, not from distance.
- A fast fading channel produces an **irreducible BER floor**: extra transmit power does not help, because the channel has already changed by the time the symbol ends.
- In practice fast fading only matters for **very low data rates**.

3.8 Rayleigh and Rician fading

Rayleigh fading channel Vs Ricean fading channel, with appropriate expressions 5 2 2+4

HOT 5/22 70 Ma, 71 Ma, 72 Ash, 73 Bh, 80 Ch

What these distributions describe

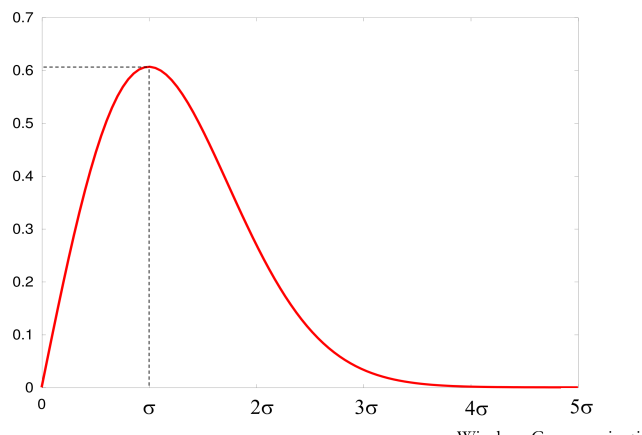
- Not *when* the signal fades, but **how the received envelope is statistically distributed**.
- Both apply to **flat fading** channels.

Rayleigh

- Applies when **all multipath components have roughly equal amplitude** and there is **no dominant LOS component**.
- Physically: mobile deep in an urban area, far from the BS, no line of sight.
- The in-phase and quadrature sums are each Gaussian, so the **envelope** is Rayleigh:

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & 0 \leq r \leq \infty \\ 0 & r < 0 \end{cases}$$

- σ = rms value of the received voltage **before** envelope detection.
- σ^2 = time-average power before envelope detection.
- **Phase** is uniformly distributed over 0 to 2π .
- Key values:
 - mean $r_{\text{mean}} = \sigma \sqrt{\pi/2} = 1.2533 \sigma$
 - median $r_{\text{median}} = 1.177 \sigma$
 - variance $\sigma_r^2 = 0.4292 \sigma^2$
- **Mean and median differ by only 0.55 dB**, which is why the median is used in practice.



Ricean

- Applies when there is a **stationary, non-fading, dominant** component, normally a LOS path, on top of the random multipath.
- Physically: microcell, indoor pico-cell, or a mobile with clear sight of the BS.
- The random components add onto a fixed phasor, so:

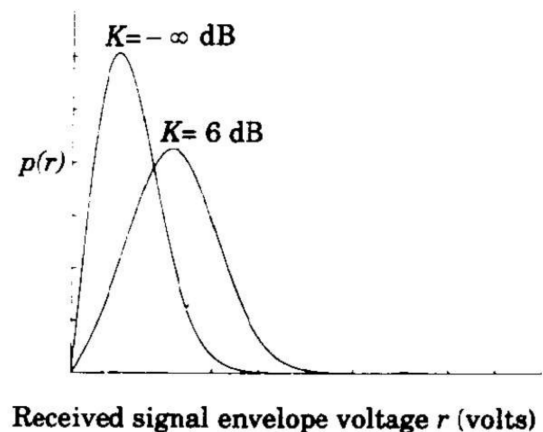
$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + A^2}{2\sigma^2}\right) I_0\left(\frac{Ar}{\sigma^2}\right) & A \geq 0, r \geq 0 \\ 0 & r < 0 \end{cases}$$

- A = peak amplitude of the **dominant** signal.
- $I_0(\cdot)$ = modified Bessel function of the first kind, zeroth order.
- **Ricean K -factor**

$$K = \frac{A^2}{2\sigma^2}, \quad K [\text{dB}] = 10 \log \frac{A^2}{2\sigma^2}$$

= ratio of **deterministic power** to **multipath power**.

- $A \rightarrow 0$, **ie $K \rightarrow -\infty$ dB: Ricean degenerates into Rayleigh**.
- $K \gg 1$: the distribution tends to **Gaussian** about the mean, ie almost no fading.



$K = -\infty$ dB is the Rayleigh case; $K = 6$ dB shows a dominant component pulling the distribution towards Gaussian.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
WLAN	wireless local area network	WMAN	wireless metropolitan area network
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM

evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.